Learning Waypoints

A unit planning tool for Year 7 Science — Pilot onboarding

Tool: https://learning-waypoints.streamlit.app

Why this tool exists

When multiple classes are teaching the same unit, they rarely move at the same pace. Some classes reach new ideas quickly, while others need more time to consolidate. Over time, this creates drift: classes are no longer at the same point in the curriculum.

Traditional responses often make this worse. Faster classes are moved on to new content, while slower classes repeat or simplify tasks. This breaks alignment, makes common assessment harder to justify, and removes a shared reference point for planning.

Learning Waypoints was designed to solve this specific problem: how to manage differences in learning speed while keeping all classes on the same conceptual pathway.

Streamed and unstreamed classes

This problem exists in both streamed and unstreamed settings, but it shows up differently. In streamed classes, pace differences often appear early, and acceleration pressure can quickly push faster groups into new curriculum content. In unstreamed classes, pace divergence tends to emerge gradually as units progress, leading teachers to make informal adjustments that accumulate over time.

Learning Waypoints is designed to work in both contexts. It does not replace streaming decisions or remove classroom heterogeneity. Instead, it provides a common planning structure that allows classes with different starting points or learning speeds to remain aligned conceptually, even when their journeys through the unit feel different.

The core idea

Every unit has a sequence of ideas students must pass through to reach the curriculum goal. Learning Waypoints calls these ideas waypoints. At each waypoint, classes may move at different speeds.

Rather than advancing faster classes to new concepts, the model holds all classes at the same waypoint and varies how deeply the concept is explored. Some classes consolidate, while others widen understanding through additional contexts, representations, and connections.

All classes move through the same conceptual sequence and reach the same endpoint. Differences in pace are managed horizontally, not by acceleration.

In short

- All classes stay on the same conceptual sequence
- Pace differences are expected and planned for
- Faster progress leads to wider understanding, not new content
- Slower progress does not reduce conceptual expectations

The shorthand used in the tool

Y-axis: the shared conceptual sequence from prior knowledge to the unit goal.

Waypoints (nodes): key conceptual steps along that sequence.

Width (X): how broadly a concept is explored at a waypoint (Xmin, X+, X++).

Hinge concepts: waypoints where weak understanding causes downstream problems.

Pace profile: a temporary, classlevel description of how quickly a cohort moves through new ideas and how much consolidation it typically needs before moving on. Pace profile reflects learning speed, not ability, and can change over time.

What "width" means in practice

At the same waypoint, one class may work with a single representation and familiar example, while another applies the same idea across multiple contexts, compares representations, or examines common misconceptions. Both classes are learning the same concept; the difference is how much cognitive demand is built in while they are there.

Where the waypoint sequence comes from

Waypoints are not lesson activities and are not generated at random. They represent the conceptual steps students must move through to get from prior learning to the curriculum goal.

Sequences are built from ACARA content descriptions and elaborations, known conceptual dependencies, and common breakdown points that affect later learning. In the current pilot, waypoint sequences are pre-built and can be reviewed but not edited.

Assessment alignment

Assessment is designed to support shared pacing. A core layer (around 60%) assesses minimum conceptual understanding and remains accessible regardless of class pace.

Extended layers reward broader and deeper understanding of the same concepts without introducing new content. This allows classes moving at different speeds to sit the same assessment fairly.

What the tool does

Learning Waypoints guides users through curriculum selection, waypoint review, assessment planning, and lesson prompting. Outputs help teachers manage pace differences while keeping classes aligned.

The full process typically takes 15–20 minutes for an initial unit. Outputs are a starting point, not scripts.

What this tool is not

Learning Waypoints is not intended to accelerate some classes to new curriculum content, create separate extension units, label ability, or lower expectations for slower-moving classes. All classes remain on the same conceptual pathway at all times.

How to use it well

Start with a real class you teach. Use classroom evidence to judge pace and consolidation needs. Adjust width decisions as the unit unfolds, while keeping the conceptual sequence fixed. The goal is alignment, not uniformity.

Scope and limits

Learning Waypoints is designed for subjects and units where learning can be organised as a shared conceptual progression with common endpoints. It is most effective in conceptually cumulative domains taught with aligned curriculum and assessment across multiple classes. The model is not intended for fully openended, nonsequential, or highly personalised learning contexts where progression is individual, emergent, or multidimensional.

Possible extensions

A natural extension would be to explore whether the same model applies to units organised around shared process progression. Examples might include a design cycle unit, a science investigation unit, or a text analysis unit built around a shared text. Feedback from teachers working in these contexts is welcome.

	Concept progression	Process progression
Y-axis	Shared conceptual sequence	Shared process sequence
Waypoints	Conceptual steps	Process stages
Width	How broadly a concept is explored	How broadly a process stage is explored
Hinge	Weak concept causes downstream problems	Weak process thinking causes downstream problems

Pilot note

This is an early pilot version. Rough edges are expected. Please focus on whether this approach helps you manage pace differences across classes while preserving conceptual coherence.

Feedback prompts

- Does the waypoint sequence help keep classes aligned despite pace differences?
- Does widening understanding feel like a workable alternative to acceleration?
- Would this make planning and assessment across multiple classes easier?
- What would you want the tool to do next?